Predicting Unstable Handovers Using Machine Learning in LTE/5G Networks

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***Abstract*—Frequent and unnecessary handovers are a major challenge in dense cellular networks because they increase signaling overhead and reduce the benefit of mobility management. This project studies unstable handover prediction using a machine learning pipeline built on LTE simulation and mobility traces. The main target is defined as unstable_15 = reversal_15 OR short_dwell_15. We evaluate Logistic Regression, Explainable Boosting Machine (EBM), LightGBM, and XGBoost by grouping train-test splits by UE and performing offline gating analysis. Results show that boosted-tree models perform better than the linear baseline on ranking metrics, and LightGBM has the strongest balanced threshold tradeoff for offline gating. The study shows that pre-handover measurements can be used to identify a notable amount of unstable handover events. It also highlights the difficulty of the overall task and how sensitive gating performance is to model choice and threshold selection.**

***Index Terms*—Unstable handover, supervised machine learning, 5G/LTE, offline gating, sensitivity analysis**

I. Introduction

Cellular handover is important for maintaining service continuity when the user moves around, sometimes between coverage areas as well. But, in dense deployments, due to smaller base station coverage areas, target base station changes rapidly which can trigger unnecessary handovers. This issue is known as the ping-pong effect; these occur when user equipment (UE) switches back and forth between cells quickly. These events increase control overhead and can degrade quality of service (QoS). Traditional handover logic is usually based on radio KPIs such as signal quality of serving and neighbor cells, hysteresis, and time-to-trigger parameters. These metrics are very effective for basic mobility management, i.e., predicting if a handover should take place, but they do not predict whether an actual handover becomes unstable, thus leading to a ping-pong situation. This motivates more outcome-oriented research in which the model predicts post-handover instability from pre-handover observations. Unlike prior work that predicts trigger events, target cells, or handover failures, this paper predicts whether an executed handover will later become unstable.

The main idea of this project is to move from KPI trigger to outcome prediction, i.e., instead of predicting whether a KPI-style trigger condition holds, the system predicts whether an actual handover later becomes unstable. In this approach, we create a per-second dataset built with ns-3 [10] using SUMO [11] mobility and extract the actual handover events from the change in serving cell ID. We then create event-level labels from dwell time and reversal behavior, and evaluate multiple machine learning models through gating tradeoffs. The main contributions of the project are: (i) generation of a dataset using ns-3 simulation with SUMO mobility for collecting radio and mobility KPIs used for training; (ii) creation of an outcome-based instability label using short dwell time and rapid reversal; (iii) comparison of linear, interpretable nonlinear, and boosted-tree models; and (iv) threshold-based offline gating analysis that measures the tradeoff between unstable-handover reduction and stable-handover loss.

The remainder of the paper is organized as follows. Section II presents related work using machine learning techniques for handover optimization. Section III gives an overview of handover architecture in 5G/LTE networks. Section IV presents the system model and overview of dataset generation. Section V shows the performance evaluation of the applied techniques. Section VI concludes the work and outlines future scope.

II. Related Work

In the literature, there has been extensive research on handover optimization in 5G/LTE networks. Recent works can be divided into four categories: (i) dwell-time based handover control; (ii) machine-learning based handover decision models; (iii) handover failure prediction; and (iv) enhancements in architecture-level mobility, such as conditional handovers or SDN-assisted coordination. All these studies share a common motivation: dense deployments lead to more frequent threshold-based handovers, short dwell time, and unnecessary signaling overhead.

In the first category, [3] uses a combination of dwell-time prediction and the TOPSIS algorithm with metrics like RSS, throughput, cost, delay, and SNIR to reduce redundant handovers. Similarly, [8] uses expected dwell-time duration for selecting cells to control or reduce the number of handover decisions. These studies show the importance of dwell time as a target. This work uses dwell time to create a final unstable target that captures the instability of the executed handover decision.

In the second group of studies, [2] uses Logistic Regression on a simulated dataset with radio KPIs such as RSRP, SINR, RSSI, and distance to create a trigger label that helps in predicting handovers and thus reducing them. This work is an important baseline to understand how a simple linear model can help improve mobility management. Our work is in the same direction but differs in three ways: (i) the target is not a trigger decision but post-handover instability; (ii) the label is a combination of reversal logic and short dwell time; and (iii) we include more than one model for a broader comparison among linear, interpretable nonlinear, and boosted-tree models.

Another line of work focuses on handover failure prediction. [1] predicts handover failure by treating it as a classification problem and comparing several classifiers, with XGBoost as the strongest one. They use radio features from serving and target cells as input. This work shows the importance of feature selection for machine learning-based mobility tasks. However, our target label is broader. In our case, a handover may complete successfully but might still be unstable when it remains in the target cell for a short time before moving on to another cell or back to the previous cell. This makes our label more directly connected to unnecessary handover mitigation.

A rising body of work adds mobility-aware or sequential AI into the handover procedure. References [4] and [5] use mobility information taken from the handover decision process to reduce frequent handovers in dense small-cell environments. Reference [7] reduces unnecessary resource allocation in candidate cells using a conditional handover framework that predicts target-cell lists and dwell time more efficiently. These studies highlight the importance of temporal context and prediction in improving mobility management beyond threshold rules. Unlike these studies, we intentionally formulate the problem as an offline event-level prediction task using handover traces extracted from simulation. This controlled setting lets us isolate whether pre-handover radio and mobility features contain enough signal to predict post-handover instability before moving to closed-loop deployment.

Other works such as [6] reduce inter-frequency handover delay by adding machine learning support instead of standard measurement-based selection. Reference [9] selects the best cell and reduce handovers using SDN-based coordination and optimization. These works demonstrate that handover optimization affects not only decision quality, but also latency, signaling overhead, and control-plane efficiency. Table I summarizes the main differences between representative prior work and this study.

Overall, prior work mainly focuses on trigger prediction, dwell-time-aware target selection, handover failure prediction, or architecture-level mobility improvements. In contrast, our work treats post-handover instability as the main prediction target. We define instability using short dwell time and rapid reversal, train multiple machine learning models on event-level samples, and evaluate them through offline gating tradeoffs between unstable-handover reduction and stable-handover loss.

III. Handover in 5G and LTE Architecture

Handover is an essential mobility-management procedure in cellular systems today. In LTE and 5G systems, the handover decision is based on several factors such as radio measurements, mobility-management logic, and signaling exchange between serving and target cells. Fig. 1 shows the main entities involved in the handover process and the overall message flow between the UE, source cell, target cell, and core-control elements.

A handover occurs when a UE changes from one cell to another without ending the ongoing communication session. Since the coverage area of any base station is limited, a moving UE cannot remain attached to the same base station. As the UE moves, the quality of the signal from the current base station degrades to a point where the neighboring cell becomes more suitable. In that situation, the network must decide whether to keep the current connection or transfer the call to another cell.

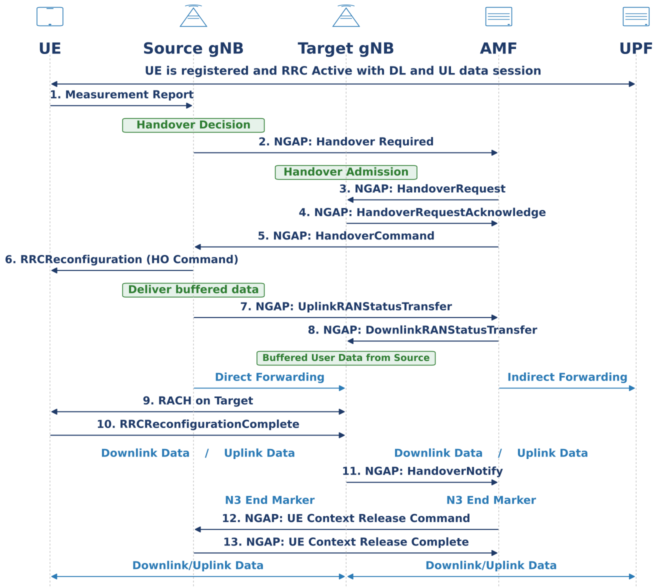

Fig. 1. 5G/LTE Handover Architecture.

Handover becomes more important in dense areas. In 5G systems, networks use many closely spaced cells to increase capacity and improve radio-resource reuse. This improves coverage but also causes overlapping regions and more frequent transfers. As a result, the number of handovers increases. There may arise cases where a UE is moved too often such that it remains with the new cell for a very short time before it is moved back to the previous cell.

Because handover affects the user experience, networks track handover-related metrics and requirements. A good handover should maintain service continuity and keep interruptions low. In dense deployments, another important concern is instability, such as short dwell time in the target cell or a quick return to the previous cell. These suggest that handover decisions triggered according to radio thresholds may have been low-value, and thus a waste of resources.

In this project, handovers are studied through an ns-3 + SUMO simulation environment using LTE mobility-management mechanisms. The main contribution of this paper is to provide a machine-learning based framework for predicting whether a handover is likely to become unstable

TABLE I
COMPARISON OF RELATED WORK

Work	Main objective	Method/Models	Main inputs	Scenario	Main result / focus	Difference from this work
[1]	Predict handover failure	SVM, DT, RF, XG-Boost, KNN, Naive Bayes	Serving/target/interferer RSRP	5G inter-frequency HO	XGBoost gives strongest failure prediction	Targets handover failure, whereas our work targets post-handover instability including short dwell and reversal.
[2]	Reduce unnecessary handovers	Logistic Regression	RSRP, SINR, RSSI, distance	5G simulation	HO reduction and improved mobility efficiency	Serves as baseline reference; our work predicts instability after handover rather than handover occurrence or trigger behavior.
[3]	Reduce unnecessary handovers	Dwell-time prediction + TOPSIS	RSS, throughput, cost, delay, BER, SNIR	5G HetNet	Improves decision quality through dwell-aware target selection	Uses multi-criteria cell selection; our work learns event-level instability risk from pre-handover features.
[4]	Mitigate frequent handovers	Mobility-aware handover control	RSRP pattern during TTT, mobility cues	Small-cell network	Reduces frequent handovers	Focuses on mobility-aware procedure design; our work uses supervised event-level prediction and offline gating evaluation.
[5]	Improve conditional handover efficiency	Multi-task deep learning	Target-BS list and dwell-time sequence	B5G/6G conditional HO	Reduces excessive reservation in CHO	Addresses conditional handover resource timing, while our work evaluates instability prediction and gating in offline form.
[6]	Reduce handover latency	Machine-learning-based target prediction	Live-network measurements	Inter-frequency HO	Reduces procedure delay	Focuses on latency reduction in the signaling procedure rather than instability of executed handovers.
[7]	Reduce unnecessary handovers in dense networks	ML-based mobility prediction	Mobility and handover context	Ultra-dense network	Reduces handover rate and instability-related effects	Close in motivation, but our work emphasizes explicit instability labels and compares linear, interpretable nonlinear, and boosted-tree models.
This work	Predict unstable handovers and suppress low-value mobility events	Logistic Regression, EBM, LightGBM, XGBoost	Radio, geometry, mobility, trigger, and temporal delta features	SUMO + ns-3, LTE mobility with 5G motivation	Offline gating tradeoff between unstable-HO reduction and stable-HO loss	Event-level instability prediction using unstable_15 = reversal_15 OR short_dwell_15.

after execution and thus can be avoided, using pre-handover radio and mobility features.

IV. SYSTEM MODEL

A. Dataset Generation

The simulation pipeline is built using ns-3 LTE and SUMO mobility. Using SUMO, a Manhattan-style mobility scenario is created. There are 50 UEs that are continuously moving using TraCI. This generated a trajectory file which was then replayed to produce a per-second radio and mobility dataset. The final dataset contains 18 features:

- **time_s**: Simulation time in seconds for the current sampled row.
- **ue_imsi**: Unique identifier of the UE (user equipment).
- **rssi_id**: Cell ID of the candidate/strongest neighbor associated with RSSI.
- **rssi**: Received signal value of the candidate cell in dBm used for comparison against the serving-side value.
- **current_id**: Cell ID of the currently serving eNB/cell for that UE at that time.
- **current_rssi**: Received signal value of the current serving cell in dBm.
- **hysteresis**: Handover hysteresis margin in dB.
- **rsrp**: Serving-cell RSRP in dBm; current_rssi == rsrp.
- **sinr_db_reported**: SINR in dB as reported from the callback/measurement interface.
- **sinr_linear**: SINR in linear scale from the UE PHY callback ReportCurrentCellRsrpSinr.
- **sinr_db_calc**: SINR in dB, calculated from sinr_linear as $10\log_{10}(\text{sinr\_linear})$.
- **distance_ue**: Distance between the UE and the relevant serving-cell position in meters.
- **speed_ue**: UE speed at the current 1-second sample, with speed_ue $= \sqrt{dx\_ue^2 + dy\_ue^2}/dt$.
- **dx_ue**: Change in UE x-position between the current and previous 1-second sample.
- **dy_ue**: Change in UE y-position between the current and previous 1-second sample.
- **trigger_a3**: Trigger based on candidate_rsrp_dbm > serving_rsrp_dbm + hysteresis; recorded as a pulse instead of a continuous 1 over consecutive seconds.
- **trigger_kpi**: Trigger based on threshold conditions involving rsrp, sinr_db_calc, and distance_ue with two-sample confirmation and rising-edge pulse behavior.
- **gap_db**: Difference between the candidate-cell signal and the serving-cell signal in dB, i.e., candidate_rsrp_dbm - serving_rsrp_dbm.

The final dataset was simulated for 100,000 seconds (approximately 1.20 days). For 50 UEs and 8 serving cells, there were around 5 million entries, corresponding to the observed range of time from 1.0 s to 99,999.0 s. Table II summarizes the main simulation parameters used in SUMO and ns-3. After dataset creation, several sanity checks were conducted to identify inconsistencies unrelated to the mobility-management problem.

Later, the actual handovers were extracted from changes

TABLE II
SIMULATION PARAMETERS

Parameter	Value
SUMO	
simulation begin	0
simulation end	100000
step length	1.0
speed range	10 m/s to 13.9 m/s
number of eNBs	8
ns-3	
number of UEs	50
eNB layout	2 x 4
eNB placement rectangle	x=[175,1250], y=[230,830]
eNB height	30 m
eNB Tx power	46 dBm
UE Tx power	26 dBm
bandwidth	50 PRBs
HO algorithm	A3-Rsrp-HO-Algo
hysteresis	2.0 dB (std)
Time-To-Trigger (TTT)	160 ms
dataset sampling period	1.0 s

in serving cell ID into another `ho_events.csv`. For each event, the extracted fields were UE identity, handover time, source cell, destination cell, dwell time in the destination cell, and helper labels. The dwell time distribution for the dataset was highly skewed, with a minimum of 11 s, maximum of 20,338 s, and a median of 34 s.

The pre-handover features were attached to each handover event in the event-level training table `ho_train.csv`. The final training table contained pre-handover radio, trigger, and mobility information. The positive class was rare.

B. Event-Label Formation

The final target is defined as unstable_15 = reversal_15 OR short_dwell_15, where short_dwell_15 is equal to 1 when dwell_s $\leq$ 15 s and reversal_15 is equal to 1 when a handover from A to B and then from B to A happens again within 15 s. This captures two signs of instability: short stay in the target cell and short-term reversal. This outcome-based label is more meaningful than trigger prediction because it is not condition-dependent; instead, it identifies whether the executed handover later becomes unstable.

To determine a suitable instability window for the target label, we experimented with many values. We discovered that strict short-window definitions were too sparse in this case. For example, reversal counts at 5 s and 10 s were zero. This showed that 5 s and 10 s were too restrictive to capture meaningful reversal events in our setting. We therefore selected a 15 s window because it remains short enough to represent rapid post-handover instability while still providing enough positive events for meaningful modeling and evaluation. We also observed that, in the final dataset, short_dwell_15 is a superset of reversal_15, which further supports this definition.

C. Prediction Features and Models

Each event-level sample contains pre-handover information describing the network state before the event, including signal quality, serving-target SINR gap, relative geometry, sampled trigger state, and motion information. The feature pool includes pre_rsrp, pre_sinr_db_calc, pre_distance_ue, pre_gap_db, pre_hysteresis, pre_speed_ue, pre_trigger_a3, pre_trigger_kpi, pre_dx_ue, and pre_dy_ue. For the boosted-tree models, short-term temporal features were also included, namely delta_gap_1s and delta_sinr_1s. These capture whether gap_db or SINR was improving or deteriorating before the handover happened. In addition, 3-second summary features avg_gap_last3, avg_sinr_last3, gap_slope_last3, and sinr_slope_last3 were selectively added to LightGBM to provide more temporal context.

TABLE III
MAIN RANKING AND POINT METRICS ON THE 100k DATASET

Model	PR-AUC	ROC-AUC	Precision @0.5	Recall @0.5	F1 @0.5
Logistic Regression	0.1207	0.8570	0.0843	0.8035	0.1526
EBM	0.1815	0.8892	0.0000	0.0000	0.0000
LightGBM	0.1967	0.8889	0.1281	0.7018	0.2167
XGBoost	0.1912	0.8930	0.1146	0.8421	0.2018

V. PERFORMANCE EVALUATION

To build a stricter evaluation protocol, all models were evaluated on data with train-test splits grouped by ue_imsi so there was no leakage across training and testing. The positive class is rare, with around 2.8% in the entire dataset, so PR-AUC is a better metric than accuracy. This setting is harder than trigger prediction because the target is a rare post-handover outcome.

The positive class is rare, so we addressed class imbalance in two ways. First, we emphasized PR-AUC and event-level gating tradeoffs instead of raw accuracy. Second, we used imbalance-aware weighting during training and keep the test distribution unchanged. The boosted-tree models used positive-class weighting, and Logistic Regression used class-balanced weighting. We also tested partial SMOTE on the training data, but it did not improve the metrics or the operating-point tradeoffs. Therefore, we report the final results using the original data and class-weighted learning. This keeps the precision-recall results more realistic for deployment, even though it produces more conservative scores.

Table III presents the main ranking and point metrics for the evaluated models on the 100k dataset. The baseline logistic regression model is useful as a reference, but the boosted-tree models dominate in ranking quality. XGBoost provides the strongest PR-AUC and ROC-AUC, making it the best overall model for ranking unstable handovers above stable ones. However, LightGBM with short-term temporal summary features has the best overall balance between ranking and point-decision performance, and its threshold behavior is more suitable for practical gating. EBM is useful as an interpretable nonlinear model, but its score distribution has a narrow range, so a standard threshold like 0.5 does not work well for practical gating.

In the current work, gating acts as a pre-execution layer on top of standard handover logic. Once a candidate handover is identified, the model assigns it an instability risk score. The handover is executed if risk $\leq$ x and suppressed if risk $>$ x, where x is a globally fixed threshold during the offline evaluation for consistent comparison across models. The raw

TABLE IV
SUGGESTED HEADLINE POLICY OPERATING POINTS FROM THE 100K THRESHOLD SWEEPS

Model	Threshold	HO reduction %	Unstable reduction %	Stable-HO loss %	Reversal reduction %
Logistic Regression	0.7	11.02	53.01	9.43	57.58
EBM	0.1	9.13	52.98	7.86	46.81
LightGBM	0.7	8.41	47.37	7.28	40.43
LightGBM (aggressive)	0.6	11.72	60.70	10.31	55.32
XGBoost	0.7	11.08	57.54	9.73	49.47

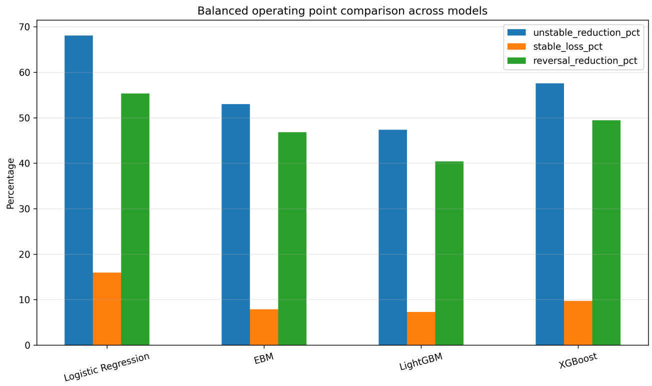

Fig. 2. Balanced operating point comparison across models.

score is not shown directly, but its effect appears in Table IV through threshold-based tradeoffs.

Table IV lists the main operating points selected from the threshold sweeps and highlights the tradeoff between unstable-handover reduction and stable-handover loss. As shown in Fig. 2 and Table IV, LightGBM with 3-second temporal summary features at threshold 0.7 gives the best overall tradeoff, achieving 47.37% unstable-handover reduction at only 7.28% stable-handover loss. This is the strongest balanced policy result in the current study. If a more aggressive policy is preferred, the same model at threshold 0.6 increases unstable-handover reduction to 60.70% at a cost of 10.31% stable-handover loss. This behavior is also consistent with the threshold trends in Figs. 4 and 5 and the full policy tradeoff curve in Fig. 6. XGBoost remains the strongest ranking model, while Logistic Regression serves as a simple baseline, and EBM is most useful at lower thresholds.

The figures help explain these results from different angles. Fig. 2 summarizes the selected operating points across models and compares unstable-handover reduction, stable-handover loss, and reversal reduction. Fig. 3 gives a direct baseline-versus-proposed comparison between Logistic Regression and LightGBM. Figs. 4 and 5 show how stable-handover loss and unstable-handover reduction change as the threshold varies. Fig. 6 combines both effects into a single tradeoff curve, which is the most useful view for selecting a practical operating point. Among these plots, Fig. 6 is the most important because it directly shows how much unstable-handover reduction each model can achieve for a given level of stable-handover loss.

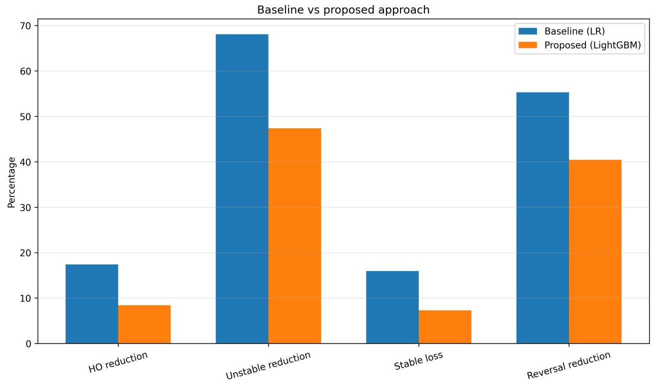

Fig. 3. Baseline vs. Proposed.

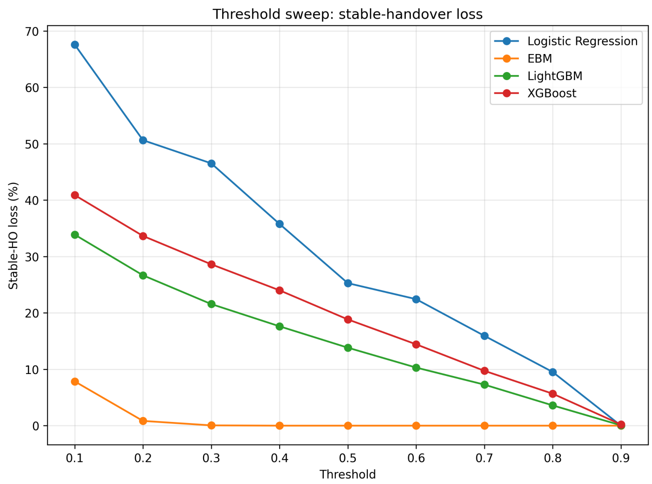

Fig. 4. Threshold vs. Stable Loss.

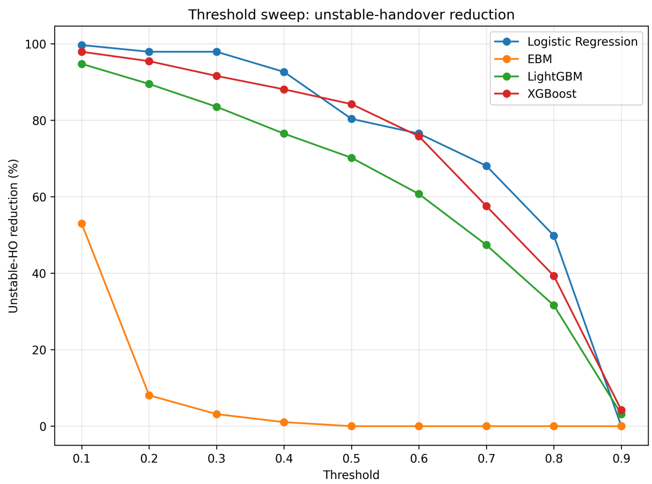

Fig. 5. Threshold vs. Unstable Reduction.

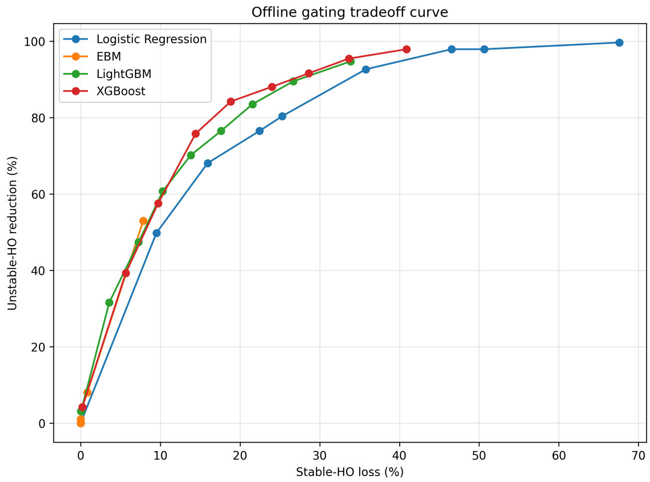

Fig. 6. Offline gating tradeoff curve.

VI. Conclusion and Future Work

We presented an event-level machine learning framework for predicting unstable handovers using data generated with ns-3 and SUMO. Instead of using a trigger as the final target, we predicted post-handover instability defined by short dwell time and rapid reversal behavior. The results show that pre-handover radio and mobility features can identify a meaningful fraction of unstable handover events. Among the four models we tested, the boosted-tree models gave the strongest results. XGBoost achieved the best ranking performance, and LightGBM gave the most practical threshold tradeoff for offline gating.

This study establishes a controlled baseline for post-handover instability prediction from pre-handover features. A natural next step is to move this offline pipeline into an online simulator loop, where the model is queried before handover execution and the mobility sequence changes accordingly. We can also add richer temporal context, such as multi-second summary features or short sequence models. We can also test different dwell-time and reversal windows to study how sensitive the results are to the label definition. This would help us understand not only ranking performance but also the impact of the model in a closed-loop mobility-management setting.

Although we built this study on simulated LTE mobility traces, the workflow can be used in a realistic deployment setting. We could place the model near the handover decision function and query it just before execution using measurements that the network collects, such as serving-cell quality, target-cell gap, geometry-related context, and short-term mobility indicators. The model would then output a risk score showing whether the candidate handover is likely to become unstable. An operator could choose a threshold based on system goals, for example reducing more unstable handovers or protecting more stable ones. But real deployment would have constraints that simulation does not fully capture, such as vendor-specific measurement behavior, changing traffic and interference patterns, limited feature availability, and strict latency requirements inside the mobility-control loop. Even without closed-loop online execution, the present results provide useful information for understanding which pre-handover features and model families are most promising for instability-aware mobility control.

This project has two important limitations. First, we used simulated data, so the mobility and handover behavior may not completely match real networks. A natural next step is to validate the approach on real network traces. Second, unstable events were rare and made up only about 2.8% of the entire dataset. In future work, we can explore additional ways to address this imbalance, including generative methods or synthetic-data construction, and then test whether those approaches improve offline evaluation without reducing realism.

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